

Optimization for Data Science

Lecture 2: Mathematical Preliminaries

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Where This Lecture Fits

$x \rightarrow$ Probabilistic Optimization
approaches...

Context in the Course:

- Lecture 1 introduced optimization in data science: what is optimization, why we care.
- **Now:** We dive into essential math concepts:
 - ✓ Vectors, norms *vector norms: $l_1, l_2 \dots l_n$ norm??*
 - ✓ Gradients, Hessians *∇, ∇^2, ∇*
 - • Taylor expansions ←
 - • Eigenvalues ←
- These fundamentals power many techniques:
 - • Gradient descent, Newton's method
 - • Support Vector Machines / SVR
 - • PCA (via eigenvalues), etc.

✓ Deterministic Optimization
+
✓ stochastic evolutionary algorithms...

Vectors: Basic Definition (1/2)

multi variate optimization: Objective f on: $x_1, x_2, \dots, x_n$

all at the same time...

→ profit safety reliability

Notation:

- A vector $\mathbf{x} \in \mathbb{R}^d$ has d components: $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$
- Bold font often denotes vectors, e.g. $\mathbf{w}$ for parameters, $\mathbf{x}$ for data point.

$\begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}$

n - number of decision variables...

In Data Science:

- A single data sample's features (like $(x_1, \dots, x_d)$)
- A model's coefficients, e.g. $\mathbf{w} = (w_1, \dots, w_d)$.

$$f: x_1 + x_2 + x_3 + x_4$$

$$f: \ln(x_1) + \ln(x_2 x_4) + x_3^3$$

→ Common sense

$$f(x_1, x_2, \dots, x_n)$$

one objective f on

n - decision variables

& "convexity"

Vectors: Basic Definition (2/2)

→ decision variables
→ parameters

$$\gamma = \vec{x}^T \vec{w}$$
$$Y = \alpha X + \beta \quad \gamma = [\gamma_1, \gamma_2]^T \quad \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$
$$\vec{\theta} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$
$$\vec{\theta} = [\alpha \ \beta]^T$$

Row vs. Column:

- $\mathbf{x}^T$ is row vector; $\mathbf{x}$ is column vector.

Example:

$$Y = \gamma_1 \theta_1 + \gamma_2 \theta_2 \quad \vec{w} = \begin{pmatrix} 2.5 \\ -0.8 \end{pmatrix} \quad (2D \text{ vector}).$$

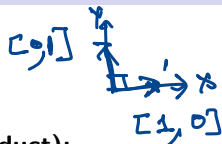
Why We Need This:

- We'll frequently do $\boxed{\mathbf{x}^T \mathbf{w}}$ to compute predictions in linear regression, logistic regression, etc.

{ "vector operations": mathematical convenience! }

Dot Product (1/2)

Definition (Dot Product):



$$[0, 1] \cdot [1, 0] = 0$$

$$\Rightarrow \boxed{\mathbf{x}^T \mathbf{y}} = \sum_{i=1}^d x_i y_i$$

$$\begin{aligned} &\rightarrow \mathbf{x}^T \boldsymbol{\theta} \\ \mathbf{x} &= [x_1, x_2]^T \\ \boldsymbol{\theta} &= \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} \end{aligned}$$

Interpretation:

- If vectors are aligned, dot product is large positive.
- If perpendicular, dot product is zero.
- If opposite directions, negative.

$$\mathbf{x}^T = [x_1, x_2] \quad \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$

$$\mathbf{x}^T = x_1 \theta_1 + x_2 \theta_2$$

Dot Product (2/2)

Geometric Angle:

$$\mathbf{x}^T \mathbf{y} = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos(\theta),$$

where θ is angle between them.

In Data Science:

- Similarity measure (cosine similarity for text vectors).
- Basic operation in linear regression: $\mathbf{w}^T \mathbf{x}$.



l-2 norm!!
|| ||₂
norm?
②

Vector Norms: Measuring Size (1/2)

ℓ_2 -Norm (Euclidean):

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + \dots + x_d^2}$$

$$\|\mathbf{x}\|_2 = \sqrt{(x_1 - 0)^2 + (x_2 - 0)^2 + \dots + (x_d - 0)^2}$$
$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Meaning: ordinary distance from origin.

ℓ_1 -Norm:

$$\|\mathbf{x}\|_1 = \sum_{i=1}^d |x_i|$$

Meaning: sum of absolute values.

sum of absolute values

$$\|\mathbf{x}\|_n = \left(x_1^2 + x_2^2 + x_3^2 + x_4^2 \right)^{1/n}$$

Vector Norms: Measuring Size (2/2)

ℓ_∞ -Norm:

$$\|\mathbf{x}\|_\infty = \max_i |x_i|.$$

Why So Many Norms?

- ℓ_2 norm: typical geometry (Euclidean).
- ℓ_1 norm: used for sparsity (LASSO).
- ℓ_∞ bounding maximum coordinate, etc.

ℓ_n

-??

Example: 2D Vectors

Let: $\mathbf{x} = (2, -1)^T$, $\mathbf{y} = (3, 2)^T$.

$$\# \mathbf{x}^T \mathbf{y} = (2)(3) + (-1)(2) = 6 - 2 = 4. \quad \boxed{\equiv}$$

$$\# \|\mathbf{x}\|_2 = \sqrt{2^2 + (-1)^2} = \sqrt{5}, \quad \# \|\mathbf{y}\|_2 = \sqrt{3^2 + 2^2} = \sqrt{13}.$$

Interpretation:

- Dot product = 4 $\Rightarrow$ moderate alignment.
- Norms = lengths ≈ 2.236 and ≈ 3.606 .

$$\boxed{\equiv} \mathbf{x}^T \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \underline{\underline{\cos \theta}}$$

↓
length!

Partial Derivatives (1/2)

stationary point

$\nabla f = 0$: necessary
but
not sufficient !!

1D vs. Multi-D:

- In 1D, derivative $f'(x)$ tells slope at point x .
- In $\mathbb{R}^d$, we have d partial derivatives.

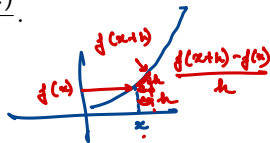
$$\frac{df}{dx} = \frac{f(x+h) - f(x)}{h}$$

Notation:

$$\frac{\partial f}{\partial x_j}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\dots, x_j + h, \dots) - f(\dots, x_j, \dots)}{h}$$

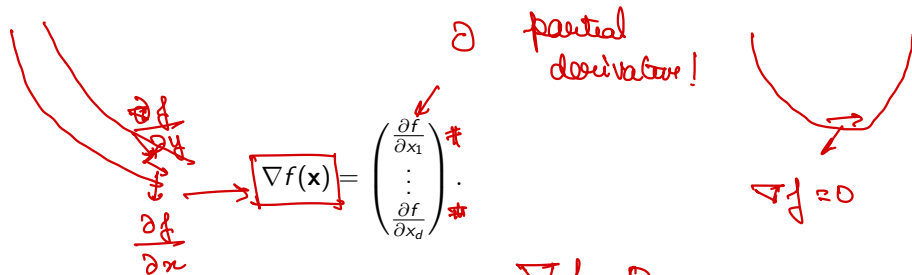
n -dimensions ↑

$$\frac{f(x+h, y+h) - f(x, y)}{h}$$



Partial Derivatives (2/2)

Gradient:



Why:

- $\nabla f(\mathbf{x})$ points toward steepest ascent.
- For **descent**, we go $-\nabla f(\mathbf{x})$.
- Setting $\nabla f = \mathbf{0}$ finds stationary points (min, max, or saddle).

$$\nabla f = 0$$
$$\frac{\partial f}{\partial x_1} = 0$$
$$\frac{\partial f}{\partial x_2} = 0$$
$$\vdots$$

Gradient Example: 2D Polynomial

Function:

$$f(x_1, x_2) = x_1^2 + 3x_2 + 2x_1x_2.$$

$$\frac{\partial f}{\partial x_1} = 2x_1 + 2x_2, \quad \frac{\partial f}{\partial x_2} = 3 + 2x_1.$$

$$\nabla f(x_1, x_2) = \begin{pmatrix} 2x_1 + 2x_2 \\ 3 + 2x_1 \end{pmatrix}.$$

Setting $\nabla f = \mathbf{0}$ might find optimum or stationary point.

$$x_2 = -\frac{3}{2}$$



$$\frac{\partial f}{\partial x_1}$$

$$\frac{\partial f}{\partial x_2}$$

$$\begin{aligned} \frac{\partial f}{\partial x_1} &= 0 \\ \frac{\partial f}{\partial x_2} &= 0 \end{aligned}$$

$$\begin{aligned} 2x_1 + 2x_2 &= 0 \\ 3 + 2x_1 &= 0 \\ x_2 &= -3/2 \end{aligned}$$

$$\begin{aligned} (x_1, x_2) &= (3, 3) \Rightarrow \frac{\partial f}{\partial x_1} = 12 \\ (x_1, x_2) &= (3, -3) \Rightarrow \frac{\partial f}{\partial x_2} = 9 \end{aligned}$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The Hessian (1/3): Taylor Expansion Connection

Second-Order Taylor (Multivariate):

$$f(\mathbf{x}_0 + \Delta \mathbf{x}) \approx f(\mathbf{x}_0) + \underline{\underline{\nabla f(\mathbf{x}_0)}}^T \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^T (\nabla^2 f(\mathbf{x}_0)) \Delta \mathbf{x}.$$

Hessian: $\nabla^2 f(\mathbf{x}_0)$

- Matrix of second partial derivatives.
- Captures *how* curvature changes near $\mathbf{x}_0$.

Usage:

- Newton's Method: we minimize this local "quadratic approximation" each iteration. $\Delta \mathbf{x} = \mathbf{h}$
- If $\nabla^2 f(\mathbf{x}_0)$ is *positive definite*, that local model has a unique minimum near $\mathbf{x}_0$.

"curvature"



$$f(\mathbf{x}_0 + \Delta \mathbf{x}) = f(\mathbf{x}_0) + \Delta \mathbf{x} \underline{\underline{\nabla f(\mathbf{x}_0)}}^T$$
$$\nabla f(\mathbf{x}) = \frac{f(\mathbf{x} + \Delta \mathbf{x}) - f(\mathbf{x})}{\Delta \mathbf{x}}$$

$$\nabla f(\mathbf{x}) \approx \frac{f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x})}{\mathbf{h}}$$

Solution

univariate

$$f(x) = x^3$$

$$f''(x) < 0$$

$$f(x+h) = f(x) + \frac{h}{1!} f'(x) + \frac{h^2}{2!} \boxed{f''(x)} \quad \text{Taylor expansion!}$$

$$f'(x) = 3x^2 = 0$$

$$f'(x) = 0$$

stationary

$$f''(x) = 6x = 0$$

$$f'''(x) = 6 \quad \text{U}$$

If $f(x)$ is maxima

$$f(x) \geq f(x+h)$$

$$f(x) \geq f(x-h)$$

$$\begin{matrix} h^3 \\ h < 0 \\ -ve \end{matrix}$$

$$\begin{matrix} h^3 \\ h > 0 \\ +ve \end{matrix}$$

necessary

but not sufficient

$$f'''(x) > 0$$

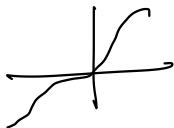
what is sufficient condition for

$f(x)$ to be maxima??

$$f''(x) \leq 0$$

$$f(x+h) = f(x) + \frac{h^2}{2!} f''(x)$$

$$\frac{f(x+h) - f(x)}{h^2} = \frac{f''(x)}{2!} < 0$$



The Hessian (2/3): Definition and Shapes

Definition (Matrix of second partials):

maxima $\nabla^2 f < 0$ minima $\nabla^2 f > 0$ $\nabla^2 f(\mathbf{x}) =$ Hessian matrix!

$$\begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1} & \dots & \frac{\partial^2 f}{\partial x_d^2} \end{pmatrix} \cdot \text{matrix!}$$

Curvature Info:

- **Positive definite** $\Rightarrow$ local minimum region.
- **Negative definite** $\Rightarrow$ local maximum region.
- **Indefinite** $\Rightarrow$ likely a saddle region.

Visual:

- In 2D, positive definite Hessian $\Rightarrow$ "bowl shape."
- Negative definite Hessian $\Rightarrow$ "upside-down bowl."
- Mixed signs $\Rightarrow$ "saddle" shape.

$$f(\vec{x} + \vec{h}) = f(\vec{x}) + \vec{h}^T \nabla f(\vec{x}) + \frac{1}{2} \vec{h}^T \nabla^2 f \vec{h} + \dots$$

Solution

multi-
variate

~~"the matrix"~~
 (A) +ve : all values
 || || > 0
 (B)
 (C) $\lambda > 0$

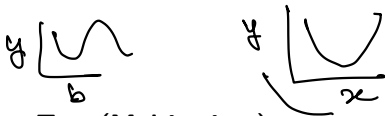
-ve matrix
 (A) -ve : All values
 || || < 0
 (B)
 $\lambda < 0$

univariate

6
 All $\lambda > 0$

-6
 $\left\{ \begin{array}{l} \lambda^2 > 0 \\ \lambda^2 < 0 \end{array} \right\}$
 → positive definiteness
 → negative definiteness

The Hessian (3/3): Sufficiency for Local Min/Max



Second Derivative Test (Multivariate):

- If $\nabla f(\mathbf{x}_0) = \mathbf{0}$ (stationary point) and $\nabla^2 f(\mathbf{x}_0)$ is:
 - **Positive definite:** $\mathbf{x}_0$ is a local minimum.
 - **Negative definite:** $\mathbf{x}_0$ is a local maximum.
 - **Indefinite:** $\mathbf{x}_0$ is a saddle point.

Data Science Example:

- In convex losses (e.g. MSE in linear regression), Hessian is positive semidefinite, so any stationary point is global min.
- In nonconvex (like deep nets), Hessian can be indefinite $\Rightarrow$ multiple minima, saddle points.

"parameters"

$$y = (ax - b)^2$$

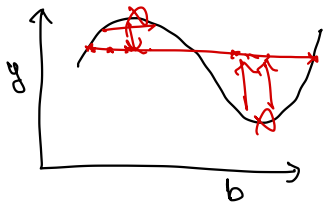
$$y = mx + c$$

$$\sum_{i=1}^n (mx_i + c - y_i)^2$$

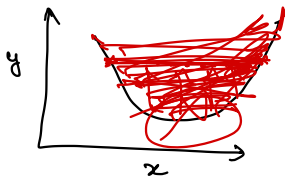
Convex in parameters

$$x^2 + c^2 + xc$$

Solution



← non-Convex



← Convex

Quadratic Functions (Extended) (1/2)

Why Quadratics Matter:

- **Least Squares:** $f(\mathbf{x}) = \|\mathbf{X}\mathbf{x} - \mathbf{y}\|^2$ is a *quadratic* in $\mathbf{x}$.
- **Local Approx (Newton):** Even if f is not globally quadratic, near a minimum it *behaves like* a quadratic if f is smooth.

Formally:

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{b}^T \mathbf{x} + \text{const.}$$

- $\mathbf{A}$ is a symmetric matrix.
- Minimizer solves $\mathbf{A}\mathbf{x} = \mathbf{b}$ if $\mathbf{A}$ is invertible.

Quadratic Functions (Extended) (2/2)

Hessian in Quadratic:

$$\nabla^2 f(\mathbf{x}) = A \quad (\text{constant}).$$

- If A is *positive definite* $\Rightarrow$ unique global min.
- If A has negative/zero eigenvalues $\Rightarrow$ not strictly convex $\Rightarrow$ possibly no unique solution.

Small Example:

- In 2D:

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

Then $\nabla^2 f(\mathbf{x}) = A$, so f is “bowl-shaped.” Minimizer at $A\mathbf{x} = \mathbf{b}$.

Conclusion:

- Quadratic problems are easier to solve *exactly* (no iterations needed if you invert A).
- Non-quadratic $\Rightarrow$ often rely on *iterative* methods, but still use Hessian info *locally* (Newton).

Quadratic Functions (1/2)

General Form:

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T A \mathbf{x} - \mathbf{b}^T \mathbf{x} + (\text{const.}).$$

Where A is symmetric.

- If A is positive definite $\Rightarrow f$ is convex.
- Minimizer solves $A\mathbf{x} = \mathbf{b}$.

Why:

- Many DS objectives (like least squares) are quadratic in parameters.
- Also local approximations in Newton's method $\Rightarrow$ approximate your function by a quadratic near the solution.

Quadratic Functions (2/2)

Example: $d = 2$

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T A \mathbf{x} - \mathbf{b}^T \mathbf{x}.$$

Gradient: $\nabla f(\mathbf{x}) = A\mathbf{x} - \mathbf{b}$. **Hessian:** $\nabla^2 f(\mathbf{x}) = A$.

Minimizer: Solve $A\mathbf{x} = \mathbf{b}$.

Recap: Taylor Expansion (1/2)

1D:

$$f(x_k + h) \approx f(x_k) + f'(x_k) h + \frac{1}{2} f''(x_k) h^2.$$

Multi-D:

$$f(\mathbf{x}_k + \Delta \mathbf{x}) \approx f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^T (\nabla^2 f(\mathbf{x}_k)) \Delta \mathbf{x}.$$

Recap: Taylor Expansion (2/2)

First-order:

- Ignores second derivative $\Rightarrow$ leads to gradient descent steps with a manually chosen step size α .

Second-order:

- Uses the Hessian $\Rightarrow$ Newton's method step:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - (\nabla^2 f(\mathbf{x}_k))^{-1} \nabla f(\mathbf{x}_k).$$

Tradeoff:

- Newton steps converge faster near solution but cost more (inverting Hessian).
- First-order steps cheaper but potentially slower.

Gradient Descent (Reminder)

Linear Approx (1st-order):

$$f(\mathbf{x}) \approx f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T (\mathbf{x} - \mathbf{x}_k).$$

- Minimizing this linear function $\Rightarrow$ move opposite $\nabla f(\mathbf{x}_k)$ with some scale α .

Update:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

Convex:

- If f is convex, repeated steps converge to global minimum (assuming a good α strategy).

Eigenvalues and Eigenvectors: Basics (1/2)

Definition:

$$A\mathbf{v} = \lambda \mathbf{v},$$

- A is a $d \times d$ square matrix.
- $\mathbf{v} \neq \mathbf{0}$ is an eigenvector, λ is the eigenvalue.
- If A has dimension d , you can have up to d eigenvalues (some may be complex).

Interpretation:

- $\mathbf{v}$ is a direction in which A “stretches” or “scales” the vector by λ .
- If $\lambda > 1$, that direction is expanded; if $0 < \lambda < 1$, it is shrunk.

Eigenvalues and Eigenvectors: Basics (2/2)

Real Symmetric A :

- All eigenvalues are real.
- Eigenvectors can be chosen orthonormal.
- We can write $A = Q \Lambda Q^T$, where Λ is diagonal of eigenvalues, Q has eigenvectors in columns.

Why Care in Data Science?

- **Covariance matrices** are symmetric $\implies$ real eigenvalues.
- **Hessians** of real-valued functions are symmetric; their eigenvalues tell us about local curvature (min/max/saddle).
- **PCA** relies on eigen-decomposition of $X^T X$ or the covariance matrix.

Condition Number: Why It Matters

Definition:

$$\kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)},$$

when A is symmetric positive definite (so eigenvalues are positive).

Interpretation:

- Large κ means $\lambda_{\min}$ is very small (or $\lambda_{\max}$ is very large) $\implies$ ill-conditioned system.
- In optimization: gradient-based methods converge slowly if the function's curvature differs a lot across directions.

Examples:

- If $\kappa(A) = 1$, A is perfectly isotropic (like scalar multiple of identity).
- If $\kappa(A) \gg 1$, certain directions are “steep” while others are “flat,” so steps can zigzag or slow down.

Positive Definiteness: Two 2×2 Examples (1/2)

Recall: A matrix A is positive definite if for all nonzero $\mathbf{x}$,

$$\mathbf{x}^T A \mathbf{x} > 0.$$

Example A (Positive Definite):

$$A_1 = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}.$$

- Symmetric? Yes.
- We can test or check eigenvalues:
 - $\det(A_1) = (4)(3) - (1)(1) = 12 - 1 = 11 > 0$
 - $\text{trace}(A_1) = 4 + 3 = 7 > 0$
- Both principal minors $> 0 \Rightarrow$ positive definite.

Positive Definiteness: Two 2×2 Examples (2/2)

Example B (Indefinite):

$$A_2 = \begin{pmatrix} 2 & 5 \\ 5 & -1 \end{pmatrix}.$$

- Check determinant: $2 \cdot (-1) - 5 \cdot 5 = -2 - 25 = -27 < 0 \implies$ indefinite.
- So A_2 has one positive and one negative eigenvalue $\implies$ saddle direction.

Implication:

- For A_1 : local “bowl shape” $\implies$ good for local min if A_1 is the Hessian.
- For A_2 : part of shape curves up, part curves down $\implies$ no single local min, a saddle region in that 2D.

PCA (1/3): Setting Up the Problem

Data Matrix: Suppose X is $n \times d$, with each row a sample in $\mathbb{R}^d$.

- Usually we assume each column (feature) is mean-centered (zero mean).

Goal:

- Find a lower-dimensional subspace capturing **max variance** of the data.
- E.g. project onto a 1D line that best explains data spread.

Projection:

$$\mathbf{z} = \mathbf{v}^T \mathbf{x},$$

where $\mathbf{v}$ is the direction (unit vector). We want to choose $\mathbf{v}$ that *maximizes* the variance of $\mathbf{z}$ across all samples.

PCA (2/3): The Variance Objective

Variance of Projections:

$$\text{Var}(\mathbf{z}) = \frac{1}{n} \sum_{i=1}^n (\mathbf{v}^T \mathbf{x}_i)^2 = \frac{1}{n} \sum_{i=1}^n \mathbf{v}^T (\mathbf{x}_i \mathbf{x}_i^T) \mathbf{v}.$$

Using matrix form: X is $n \times d$, so $X^T X$ is $d \times d$.

$$\text{Var}(\mathbf{z}) = \frac{1}{n} \mathbf{v}^T \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^T \right) \mathbf{v} = \mathbf{v}^T \left(\frac{1}{n} X^T X \right) \mathbf{v}.$$

Hence:

$$\max_{\|\mathbf{v}\|=1} \mathbf{v}^T \left(\frac{1}{n} X^T X \right) \mathbf{v}.$$

PCA (3/3): Eigenvalue Formulation

Max Problem:

$$\max_{\|\mathbf{v}\|=1} \mathbf{v}^T \left(\frac{1}{n} X^T X \right) \mathbf{v} \iff \text{largest eigenvalue/eigenvector of } \frac{1}{n} X^T X.$$

- The direction $\mathbf{v}$ that achieves the maximum is the **principal eigenvector**.
- The maximum value is the **top eigenvalue**.

Data Science:

- For higher dimensions, we pick top k eigenvectors to get a k -dim subspace capturing the largest “variance chunk.”
- This is the basis of **Principal Component Analysis**.

Key Takeaways (1/2)

- **Vectors, Norms:** Basic ways to represent data or parameters, measure distances/angles.
- **Gradients:** First-order derivatives telling us slope in multi-dim. - Vital for gradient descent and advanced ML algorithms.
- **Hessians:** Second-order derivatives capturing curvature; used in Newton's method for potentially faster convergence.

Key Takeaways (2/2)

- **Quadratic Functions:** Appear in least squares, local approximations, easy to solve if A is well-conditioned.
- **Taylor Expansions:** - First-order $\Rightarrow$ gradient descent - Second-order $\Rightarrow$ Newton's method
- **Eigenvalues:** - Affect Hessian's invertibility/condition - PCA uses them to find main variance directions in data.

Overall: Mastering these math tools is crucial before diving into advanced optimization methods, line search, or specialized machine learning models.

Thank You!

Questions or Comments?